

Session 8: More data visualisation and ggplot

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Overview I

- 1 Introduction
- 2 Using + in ggplot()
- 3 Scaling of axes
- 4 labs(): (Basic) labelling
- 5 theme()
- 6 Some additional resources

Introduction

Learning goals of today

- Short repetition from last week

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- Even more data viz examples

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- Even more data viz examples
- Even more ggplot options

Repetition: Adding a third variable

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- ...

Today we talk about prettyfying graphs

■ Scaling of axes

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- Scaling of axes
- labs(): Labelling

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- Scaling of axes
- labs(): Labelling
- theme(): Changing optics

Using + in ggplot()

Building graphs bit by bit with +

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- Graphs can be saved as objects
- That means they can be added to later

How to use the +

1 Create object

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- 2 assign a base graph in ggplot() to it

How to use the +

- 1 Create object
- 2 assign a base graph in ggplot() to it
- 3 Add another additional layer to it using the +

Running everything in one go (before)

```
# Scatterplot using iris data
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
            y = Sepal.Width,
            colour = Species)) +
  geom_point()
```



Using + to split up process (after)

```
# Creating iris_plot base
iris_plot <- ggplot(data = iris,
  mapping =
    aes(x = Sepal.Length,
        y = Sepal.Width,
        colour = Species))
# Creating scatterplot by adding geom_point()
iris_scatterplot <- iris_plot + geom_point()
```



Some additional information

- Name of the objects, as always, up to you

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- Can add more than one layer (more than one +)

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- Name of the objects, as always, up to you
- Can add more than one layer (more than one +)
- Useful for adding potentially multiple geoms, axis limitations and labels

Scaling of axes

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Scaling the axes

- Sometimes graphs are too messy so we want to scale
- one can either adapt data in pre-processing
- Or one can limit the axes

Scaling through pre-processing (before)

```
# Scatterplot using iris data, but only below Sepal.Length  
ggplot(data = iris,  
       mapping =  
         aes(x = Sepal.Length,  
             y = Sepal.Width,  
             colour = Species)) +  
  geom_point()
```



Quick reminder

- We learned that in the tidyverse whatever comes before the pipe becomes first argument of next function

```
# So for example ...  
iris_data <- iris |>  
  mutate(Sepal.Length = 2 * Sepal.Length)  
# ... is identical to  
iris_data <- mutate(iris, Sepal.Length = 2 * Sepal.Length)
```

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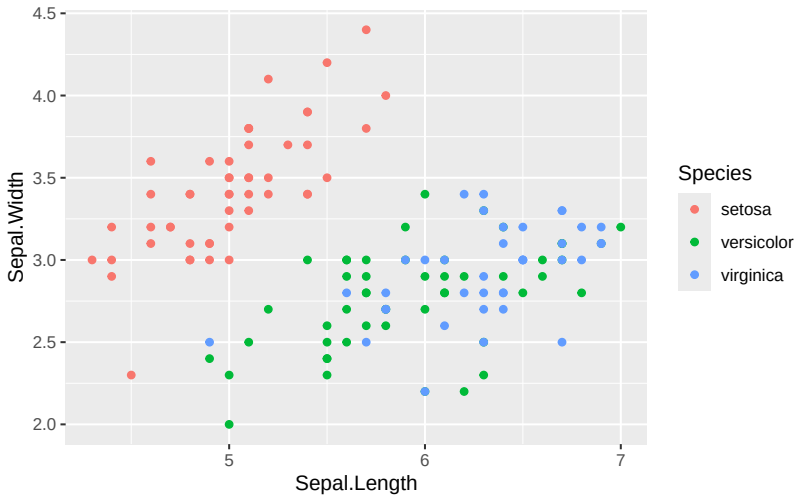
- Same holds true for ggplot()

Scaling through pre-processing (after): Code

```
# Scatterplot using iris data, but only Sepal.Length < 7
iris |> # We start off at iris data
  mutate(Sepal.Length = # then mutate() original data
    ifelse(
      Sepal.Length <= 7,
      Sepal.Length,
      NA)) |>
ggplot(mapping = # and pipe this into ggplot()
  aes(x = Sepal.Length,
      y = Sepal.Width,
      colour = Species)) +
geom_point()
```



Scaling through pre-processing (after): Image



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- May affect some statistics (like means etc.) in graph

Alternative: Limiting axes

- Alternative is adding + xlim() or + ylim() to ggplot()

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- Alternative is adding `+ xlim()` or `+ ylim()` to `ggplot()`
- Both lower and upper limits are required, but can be set to NA

Scaling through xlim() and ylim() (before)

```
# Scatterplot using iris data, but only  
# below Sepal.Length of 7  
ggplot(data = iris,  
       mapping =  
         aes(x = Sepal.Length,  
             y = Sepal.Width,  
             colour = Species)) +  
  geom_point()
```

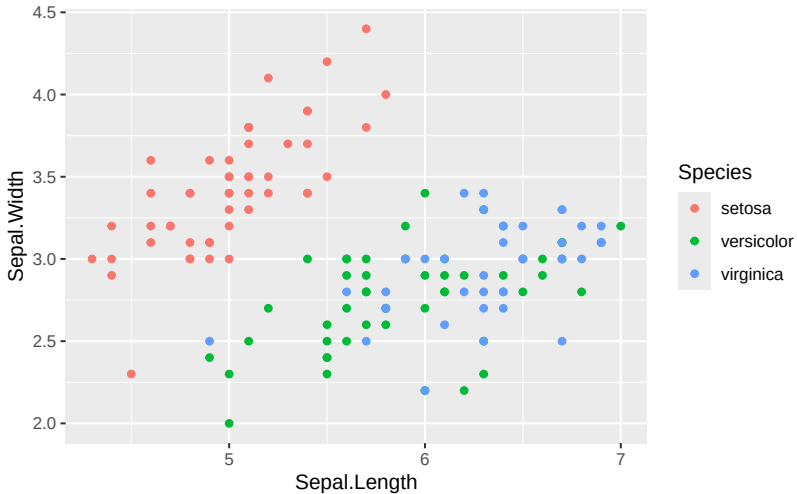


Scaling through xlim() and ylim() (after): Code

```
# Scatterplot using iris data, but only  
# below Sepal.Length of 7  
iris |>  
  ggplot(mapping =  
    aes(x = Sepal.Length,  
        y = Sepal.Width,  
        colour = Species)) +  
  geom_point() +  
  xlim(NA, 7)
```



Scaling through xlim() and ylim() (after): Image



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- Faster, but not always clear how one may limit data
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- Use coord_cartesian(xlim, ylim) for just “zooming in” while leaving underlying data intact

labs(): (Basic) labelling

Basic labelling

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- add labels using + labs(x, y, title, subtitle)

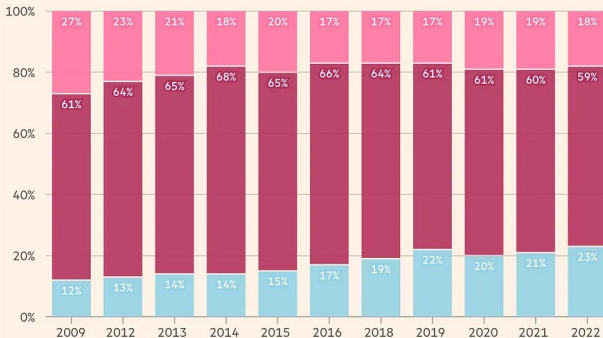
Basic labelling

- Labels are essential for understanding of graph
- add labels using + labs(x, y, title, subtitle)
- don't forget to enclose labels in “ ”

Great example: Labelling

'Quiet quitters' make up the majority of employees globally

■ Loud quitting (% actively disengaged)
■ Quiet quitting (% not engaged)
■ Thriving at work (% engaged)



Universität
Bremen

Source: www.ft.com

Creating iris_scatterplot as before

```
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iris_plot <- ggplot(data = iris,
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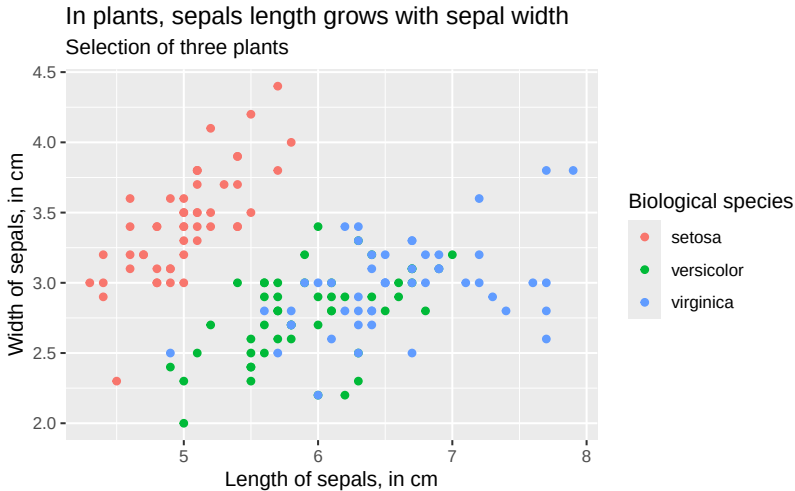


labs(): Example code

```
# Scatterplot using iris data set with labs()
iris_scatterplot + labs(
  x = "Length of sepals, in cm",
  y = "Width of sepals, in cm",
  colour = "Biological species",
  title = "In plants, sepals length grows with sepal width",
  subtitle = "Selection of three plants"
)
```



labs(): Example image



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- Re-create a graph from earlier

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- Limit the x and y variables using `xlim()` and `ylim()`

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- Re-create a graph from earlier
- Limit the x and y variables using `xlim()` and `ylim()`
- Add labels to the graph

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- These are the “vanilla” ggplot() labelling options
- Many graphs are built on ggplot(), but are using their own language
- So some of this may not be applicable, depending on the function you use

theme()

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- themes are ways to effect the colours and positioning of graph elements not related to the data or geoms
- There are a bunch of standard themes
- But also everything is customisable

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- overview here

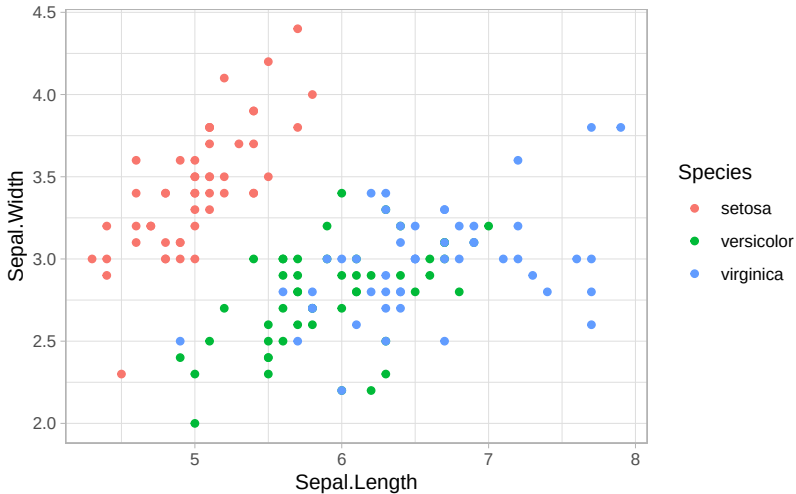
Standard theme() options

- there are a bunch of standard themes
- add them with + theme()
- overview here
- library(ggthemes) provides more cool options

theme_light(): Example code

```
# Scatterplot using iris data set with theme_light()
iris_scatterplot +
  theme_light()
```

theme_light(): Example image



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- Fonts, colours, axes lines and colours etc.

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- For more information, see here

Exercise: Create one graph with new theme

- download ggthemes

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- download ggthemes
- change the theme of one of your graphs using theme_tufte()

Exercise: Create one graph with new theme

- download ggthemes
- change the theme of one of your graphs using theme_tufte()
- What are your thoughts on this theme?

Some additional resources

Financial Times Visual Vocabulary

FT Visual Visual Vocabulary

Source: [ft.com/vocabulary](https://www.ft.com/vocabulary)

The R Graphics Cookbook

- Another great source for learning and troubleshooting

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- <https://r-graphics.org/>

Wrapping up

- Any remaining questions?

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- Working on projects